BrightSmile Dental AI Platform

Architecture review, clinical qualification gating, and real-time appointment synchronization.

< 300 ms	99.8%	100%	$34,811
VOICE RESPONSE LATENCY	INBOUND CALL COVERAGE	DETERMINISTIC RULE GATING	NET ANNUAL CLINIC SAVINGS

End-to-End System Processing Pipeline

INBOUND TO CALENDAR SETTLEMENT

STEP 01 • INBOUND
Patient Engagement
Patients connect through natural voice over WebRTC Opus audio or interactive chat, communicating concerns in everyday language.

STEP 02 • PERIMETER
Edge Guardrails
Pre-LLM sanitizers screen prompt injection attempts and medical diagnosis queries before token generation begins.

STEP 03 • GATEWAY
Session Broker
Next.js server brokers ephemeral tokens (<60s lifetime). Client audio streams peer-to-peer without exposing core credentials.

STEP 04 • DOMAIN
Clinical Engine
Authoritative TypeScript business rules match patient requests against practitioner credential tiers and operating limits.

STEP 05 • STORAGE
Atomic Commit
PostgreSQL row-level locks prevent race collisions while background adapters sync Google and Outlook calendars.

Practitioner Credential Tiers & Clinical Scope

HARD BOUNDARY RULE MATRIX

CLINICAL PROCEDURE	DURATION	DR. JUNIOR (GENERAL)	DR. SENIOR 1 & 2 (SPECIALIST)
Routine Teeth Cleaning	30 min	Permitted Standard Care	Permitted Specialist Overseen
Laser Teeth Whitening	45 min	Permitted Cosmetic	Permitted Full Scope
Root Canal Treatment	90 min	Restricted Endodontic	Permitted Primary Specialist
Wisdom Tooth Extraction	60 min	Restricted Surgical	Permitted Oral Surgery Lead
Full Mouth Reconstruction	120 min	Restricted Surgical	Permitted Surgical Lead

Deterministic Scope Enforcement
Medical authorizations are enforced deterministically in the application core (`AppointmentService`), never left to the discretion of an LLM prompt. If a patient attempts to schedule oral surgery with Dr. Junior, the scheduling engine rejects the assignment and seamlessly reroutes to an available senior doctor.

Empathetic Rerouting Dialogue
"Dr. Junior specializes in routine hygiene and whitening. For root canal therapy, our lead endodontist Dr. Senior 1 has openings this Thursday at 2:00 PM. Shall I reserve that for you?"

Operational Safety & Zero-Trust Governance

PATIENT PROTECTION STANDARDS

Adversarial Prompt Shield
Pre-parsing filters reject prompt injection patterns (such as instructions to ignore clinical rules or extract internal instructions) before triggering downstream LLM context, ensuring predictable behavior and avoiding wasted compute.

Clinical Advice Deflection
The conversational assistant strictly avoids providing diagnoses or medication prescriptions. Any inquiry regarding acute distress triggers clear guidance directing the patient to seek urgent clinical evaluation.

Privacy & Credential Hygiene
Primary API secrets and OAuth credentials stay strictly on the server. Audio is processed transiently in memory, preventing persistent capture of voice recordings while maintaining patient confidentiality.

Voice Architecture, Calendar Sync & Financials

Low-latency conversational voice mechanics, dual-calendar parity, and practice ROI modeling.

Realtime WebRTC Voice Infrastructure

SUB-300MS CONVERSATIONAL LATENCY

Latency & Interruption Profile

Traditional Chained Pipeline (STT → LLM → TTS) ~2,500 ms

Noticeable turn pauses; difficulty handling natural mid-sentence interruptions.

BrightSmile Direct WebRTC Opus Stream **< 300 ms**

Full-duplex audio; responsive interruption detection feels fluid and human.

Three-Phase Lifecycle Execution

1. Ephemeral Handshake: The browser requests a temporary session secret via `POST /api/v1/voice/session`. The backend generates a short-lived token (<60 seconds), keeping long-term keys secure on the server.

2. Direct Media Stream: WebRTC establishes a peer-to-peer connection for bidirectional Opus audio, eliminating transcription bottlenecks.

3. Server-Verified Tool Calling: When an appointment is scheduled, function calls transmit across the WebRTC DataChannel to `/api/v1/voice/tools` for authoritative database validation.

Calendar Synchronization & Atomic Integrity

PROVIDER PARITY & CONFLICT PREVENTION

Google Calendar Adapter

Executes batch FreeBusy interval queries for target practitioner calendars. Manages proactive OAuth 2.0 token refreshes in memory 5 minutes prior to expiry.

Microsoft 365 Outlook Adapter

Connects to Microsoft Graph `calendarView` via Entra ID app credentials. Recursively drains paginated busy blocks and normalizes timestamps to UTC RFC3339 format.

Dynamic Database Settings

Practitioner calendar addresses are stored in a dedicated database table, enabling clinic administrators to adjust calendar mappings through the web UI without redeploying code.

ATOMIC CONCURRENCY MODEL: 1. Fetch Database Bookings → 2. Query External FreeBusy → 3. Overlap Check (`s < end && e > start`) → 4. Postgres Row-Level Lock Settlement

Operating Economics & Practice Staffing ROI

TRANSPARENT COST ANALYSIS

INFRASTRUCTURE SERVICE	MONTHLY WORKLOAD	MONTHLY COST	SHARE
OpenAI Realtime Voice API	3,000 Spoken Minutes	**$555.00**	92.6%
OpenAI Chat Assistant (GPT-4o-mini)	15,000 Conversation Turns	**$4.05**	0.7%
Next.js Application Hosting	150,000 HTTP Requests	**$20.00**	3.3%
Managed PostgreSQL Database	10 GB SSD Storage	**$15.00**	2.5%
Google & Microsoft Cloud APIs	FreeBusy Schedule Inquiries	**$5.00**	0.8%
Total Operating Expense	**~650 Patient Appointments**	**$599.05**	**100%**

Traditional Receptionist $3,500 / mo

Covers standard business hours; missed evening calls and peak phone lines create lost booking opportunities.

BrightSmile AI Engine $599.05 / mo

24/7/365 continuous availability with immediate response, concurrent call handling, and real-time calendar locks.

ANNUAL PRACTICE SAVINGS
$34,811 / yr

83% Net Cost Reduction

VERIFICATION STATUS: 185 unit tests passing • 7 edge-case suites verified • Playwright E2E browser flows validated • Zero type errors Production Ready